

Effect of plaque brachytherapy dose and dose rate on risk for disease-related mortality in 1238 patients with choroidal melanoma

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ABSTRACT

Background Episcleral brachytherapy is the most common treatment for medium-sized choroidal melanomas. Although controversial, inadequate brachytherapy dose and dose rates have at least a hypothetical implication on patient survival.

Methods All patients who received ruthenium-106 or iodine-125 brachytherapy for choroidal melanoma at St. Erik Eye Hospital 1996 to 2016 were included (n=1238). Cox regression hazard ratios for melanoma-related mortality across deciles, quartiles and individual integers of apex radiation doses (Gy) and dose rates (Gy/hour) were calculated, adjusted for tumour size and location.

Results The average radiation dose at the tumour apex ranged from 73.0 Gy in the first decile to 108.6 Gy in the tenth. Decreasing apex dose by 1 Gy increments or by decile or quartile group was not associated with melanoma-related mortality (p>0.2). The average radiation dose rate at the tumour apex ranged from 0.5 Gy/hour in the first decile to 2.8 Gy/hour in the tenth. Similarly, decreasing apex dose rate by 1 Gy/hour increments or by decile or quartile groups was not associated with melanoma-related mortality (p>0.5).

Conclusion There are no increased hazards for choroidal melanoma-related mortality after brachytherapy with decreasing doses between 108.6 and 73.0 Gy, or with decreasing dose rates between 2.8 and 0.5 Gy/hour.

INTRODUCTION

Uveal melanoma is the most common primary intraocular malignancy in adults, with standardised incidence rates of 1.3–8.6 cases per million per year.¹ Despite long-standing efforts to improve prognosis, nearly half of all patients eventually succumb to metastatic disease.² Of all uveal melanomas, 90% arise in the choroid and only 6% and 4% in the ciliary body and iris, respectively.³ At diagnosis, half the patients have medium-sized tumours (2.5–10 mm in thickness and <15 mm in diameter).⁴ These patients are generally offered radioactive plaque brachytherapy, which conserves the eye and some degree of vision at no survival disadvantage to enucleation of the eye.⁵

Virtually all studies on brachytherapy dose and dose rate in the treatment of uveal melanoma are focused on ocular and not patient outcomes, such as local recurrence rates, treatment complications or the rate of postbrachytherapy enucleations.⁶ In the largest randomised intervention study to date,

local treatment failure including intraocular tumour growth, recurrence or extraocular extension, was observed in 10% of patients, and 13% had undergone postbrachytherapy enucleation.⁷ Similar rates were observed in one of our previous studies, where only tumours thicker than 5.5 mm were included.⁸ No increase in the number of local treatment failures has been seen with reduced radioactive dose and dose rate, within a radioactive dose (Gray, Gy) range of 62.5 to 104.0 Gy or with a high or low dose rate (Gray per hour, Gy/hour).^{6,9} In 2003, the American Brachytherapy Society recommended a dose at the prescription point, that is tumour apex, of 85 Gy.¹⁰ This guideline was updated to instead advocate an interval of 70–100 Gy in 2014.¹¹ A recommendation of a dose rate of at least 0.6 Gy/hour has remained unchanged since 2003.¹⁰

The reason for the lack of studies on patient survival in relation to brachytherapy can be found in the current understanding of the metastatic process. In 1978, Zimmerman observed a mortality peak 2–3 years after enucleation of eyes with uveal melanoma and hypothesised that enucleations may liberate tumour cells into the systemic circulation and distant organs.¹² This hypothesis has later been disproven, not least since an identical peak is observed after eye-preserving treatment.⁵ Instead, the seeding of tumour cells to the metastatic niche likely occurs very early in this disease, even before the primary tumour is diagnosed.¹³ The mortality peak can more accurately be explained by specific mutations in the tumour suppressor BAP-1, which generates early metastases, and in SF3B1, which generates later metastases.^{14,15} Consequently, neither enucleation nor plaque brachytherapy is believed to effectively prevent the development of metastases.

This study aims to investigate if a threshold for increased disease-related mortality can be identified within the clinical range of prescribed doses and dose rates at our institution. We include all patients treated with plaque brachytherapy for choroidal melanoma between 1996 and 2016, and correlate apex dose and dose rate to very long follow-up data.

MATERIALS AND METHODS

This study was approved by the regional ethical review board in Stockholm (reference number 2016/247-31/4 and amendment 2019-03485) and adhered to the tenets of the Declaration of Helsinki. All patients who received ruthenium-106

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or iodine-125 brachytherapy for choroidal melanoma at St. Erik Eye Hospital between 25 April 1996 and 16 March 2016 were considered for the study (n=1242). Retrospective data were retrieved from digitalised clinical records and from our digital Brachytherapy of Uveal Melanoma directory, including information on gender, age at diagnosis, pretreatment best-corrected visual acuity, symptom duration before presentation, type of symptoms, tumour thickness and diameter, American Joint Committee on Cancer (AJCC) T category, brachytherapy nuclide (ruthenium-106 or iodine-125), prescribed radiation dose, planned treatment duration, actual treatment duration, transpupillary thermotherapy treatment (TTT) before brachytherapy, date of death (melanoma-related, from other causes or unknown cause) and follow-up years. The date of death was retrieved from the National Mortality Registry. This registry is administered centrally and supplied with data from all healthcare institutions in the country, including classifications of cause of death. As these classifications may confuse cutaneous and uveal melanoma, we crosschecked the National Mortality Registry's data with our own clinical records of patients' metastases: if a patient had developed one or several metastases of choroidal melanoma and later died from metastasised melanoma without specified primary location, or primary location coded as skin, we considered the actual cause of death to be metastasised choroidal melanoma.

In review of the data, three patients were excluded because of errors in the data regarding apex dose and one patient due to a bilateral melanoma, yielding a total of 1238 patients eligible for the analysis. Of these, 942 were treated with ruthenium-106 and 296 with iodine-125.

Our protocol for ruthenium-106 brachytherapy, introduced in 1979, allowed for a maximum scleral dose of 1000 Gy which was increased to 1500 Gy in 1999. This means that 100 Gy at the tumour apex +1 mm can be prescribed for tumours with a thickness of up to approximately 7 mm. For iodine-125, introduced in 1999, 100 Gy was generally prescribed until 2003, after which it was lowered to 80 Gy due to concern for surrounding tissues. All ruthenium plaques were of the CCA, CCB, CCX, CCZ, COB, CIA or CIB types (Eckert & Ziegler BEBIG GmbH, Berlin, Germany). Iodine plaques are custom-made by a goldsmith in 18-carat gold alloy to replicate the CCA, CCB, CCC, CIB, COB and CCZ types, with iodine seeds (Eckert & Ziegler BEBIG GmbH) glued to the concave surface.

Ruthenium-106 is a β -ray (electron) emitter with a half-life of 373 days. It emits beta radiation with a maximum energy of 3.54 MeV and has a three times faster dose fall off than iodine-125. The plaques are used multiple times within 1 year after manufacturing, implying a gradually lower dose generated per hour (dose rate). This is compensated for by gradually longer treatment durations. Iodine-125 is a photon emitter with a half-life of 60 days. Its maximum energy is 28 keV and the low photon energy requires less shielding. It decays via electron capture to tellurium-125 which immediately decays to its ground state while emitting a 35.5 keV photon. Source specification data from the manufacturers of the radioisotopes were verified by medical physicists at the Karolinska University Hospital, Stockholm.

The brachytherapy treatment procedure was performed as we have previously described.⁹ Deviations from the prescribed dose may occur for various reasons, the most common being early or delayed access to the operating theatre, delays in anaesthesia, delayed patient transports or rescheduled surgery. Rarely, the patient experienced severe discomfort from the radioactive plaque so that its removal had to be brought forward. Such deviations from the prescribed dose, and the natural decay of

dose rates, enables the correlation with patient outcome herein presented.

Regular follow-up was scheduled at 1, 3, 6 and 12 months after therapy, and then annually or semiannually if local control had been achieved. At each visit, best-corrected visual acuity, intraocular pressure, slit-lamp examination, biomicroscopy, indirect and direct ophthalmoscopy, transillumination, ultrasonography with standardised A-scan and B-scan and fundus photography were undertaken. When tumour regression was deemed insufficient or there was tumour progression 6 months after brachytherapy or later, a second round of brachytherapy was considered. The reason for an individual patient's repeated brachytherapy or postbrachytherapy enucleation was usually described in the clinical record. At each follow-up, the visual acuity was measured on a decimal scale. Semiannual screening for liver metastases by ultrasonography or CT for 5 years after choroidal melanoma diagnosis was performed in accordance with the wishes of the patient and in collaboration with a physician at the referring hospital.

Statistical analysis

In comparisons of descriptive characteristics, patients were distributed into decile groups based on apex dose and dose rate, respectively. For a comprehensive view on the effect of incremental decreases in dose and dose rate, we analysed changes in the risk of melanoma-related mortality with incremental decreases in decile and quartile groups as well as for each decreased Gy and Gy/hour.

The actual radiation dose given was defined as:

$$\left(\frac{\text{Actual brachytherapy duration}}{\text{planned brachytherapy duration}} \right) \times \text{prescribed dose} = \text{actual dose}$$

Differences with a $p < 0.05$ were considered significant, all p -values being two-sided. The deviation of all counts from normal distribution was statistically significant, when evaluated by the Shapiro-Wilk test ($p < 0.05$). For tests of continuous variables, we therefore used the Kruskal-Wallis test. For comparisons of categorical variables, two-by-two tables and Pearson χ^2 tests were used. Kaplan-Meier analysis of disease-specific survival in relation to dose and dose rate was performed with paired deciles rendering five curves for each comparison. The Hazard Ratio (HR) for melanoma-related mortality for decreasing dose and dose rate deciles were calculated with multivariate Cox regression, adjusted for tumour size and location. HR for each decreased dose and dose rate quartile as well as for each decreased single Gy and Gy/hour were also calculated. Melanoma-related mortality was defined as death from metastatic choroidal melanoma according to data in the National Mortality Registry and/or our clinical records as described above. All statistical analyses were performed using SPSS statistics V.25 (IBM, Armonk, New York, USA).

RESULTS

Apex dose

Descriptive statistics

The average actual radiation dose at the tumour apex ranged from 73.0 Gy in the first decile to 108.6 Gy in the tenth. Tumour distance to the optic nerve head (Kruskal-Wallis $p = 0.7$) was equally distributed across the deciles. Patient characteristics and symptoms were also equally distributed across the deciles.

Tumour diameter (Kruskal-Wallis $p < 0.001$), tumour thickness (Kruskal-Wallis $p < 0.001$), AJCC T category (Pearson χ^2 $p < 0.001$) and tumour distance to the macula (Kruskal-Wallis $p < 0.001$) were not equally distributed across the deciles.

Table 1 Demographics and clinical features of patients, symptoms, tumours and treatment across deciles based on apex radiation dose

Dosedecile	1 n=123	2 n=124	3 n=124	4 n=124	5 n=124	6 n=124	7 n=124	8 n=124	9 n=124	10 n=123
Mean actual dose, Gy (SD)	73.0 (5.9)	79.8 (1.1)	87.1 (3.1)	93.7 (1.0)	96.6 (0.6)	98.4 (0.4)	99.9 (0.3)	100.8 (0.6)	103.0 (0.9)	108.6 (4.5)
Mean prescribed dose, Gy (SD)	78.4(6.8)	80.6 (3.6)	91.3 (9.7)	99.6 (2.6)	99.9 (0.48)	99.9 (0.48)	100.0 (0.0)	99.8 (1.8)	100.0 (0.32)	100.3 (4.2)
Mean patient age, years (SD)	62.1 (13.7)	63.2 (12.4)	64.3 (12.0)	65.9 (12.2)	62.4 (15.1)	63.3 (15.0)	64.7 (14.2)	63.5 (14.4)	63.9 (14.2)	62.7 (14.7)
Median symptom duration, months (IQR)	3.0 (4.0)	2.0 (2.0)	2.0 (4.0)	2.0 (5.0)	2.0 (4.0)	2.0 (1.0)	9.0 (6.0)	2.0 (3.0)	2.0 (2.0)	1.0 (2.0)
Mean visual acuity (SD)	0.4 (0.4)	0.3 (0.4)	0.4 (0.5)	0.4 (0.5)	0.5 (0.5)	0.4 (0.5)	0.5 (0.5)	0.3 (0.5)	0.4 (0.5)	0.4 (0.5)
Mean tumour diameter, mm (SD)	12.7 (3.6)	12.8 (3.4)	10.9 (3.3)	10.2 (3.4)	10.1 (3.2)	10.0 (3.4)	10.6 (3.7)	10.6 (3.3)	9.8 (3.0)	9.9 (3.1)
Mean tumour thickness, mm (SD)	7.9 (1.8)	8.4 (1.8)	5.6 (2.9)	4.1 (1.6)	4.6 (1.6)	4.5 (1.5)	4.7 (1.9)	4.6 (1.4)	4.3 (1.5)	3.8 (1.3)
Mean distance to macula, mm (SD)	4.5 (3.3)	4.5 (3.3)	3.6 (4.6)	3.2 (3.6)	3.7 (3.9)	2.8 (3.1)	3.7 (3.5)	3.2 (3.4)	3.5 (3.9)	3.5 (3.8)
Mean distance to optic disc, mm (SD)	4.4 (3.5)	4.6 (3.3)	3.9 (4.5)	3.6 (3.2)	4.0 (3.7)	3.5 (2.5)	3.7 (3.4)	3.7 (3.3)	3.3 (3.4)	3.8 (3.2)
AJCC										
T category										
1, n (%)	9 (8)	5 (4)	39 (32)	49 (42)	50 (40)	45 (39)	52 (43)	37 (31)	54 (46)	60 (52)
2, n (%)	43 (36)	35 (30)	41 (34)	54 (46)	62 (50)	61 (53)	43 (36)	72 (60)	57 (49)	53 (46)
3, n (%)	61 (51)	70 (60)	40 (33)	13 (11)	11 (9)	8 (7)	22 (18)	12 (10)	6 (5)	2 (2)
4, n (%)	7 (6)	6 (5)	1 (1)	2 (2)	1 (1)	1 (1)	4 (3)	0 (0)	0 (0)	1 (1)
Laterality										
Right, n (%)	52 (46)	55 (45)	55 (46)	61 (52)	56 (50)	60 (54)	42 (58)	62 (53)	56 (53)	47 (44)
Left, n (%)	62 (54)	68 (55)	64 (54)	56 (48)	56 (50)	51 (46)	30 (42)	54 (47)	49 (47)	61 (56)
Gender										
Male, n (%)	59 (48)	63 (51)	65 (52)	58 (47)	51 (41)	67 (54)	64 (52)	59 (48)	63 (51)	73 (59)
Female, n (%)	63 (52)	60 (49)	59 (48)	66 (53)	72 (59)	57 (46)	60 (48)	65 (52)	61 (49)	50 (41)
Pain before treatment										
Yes, n (%)	1 (1)	0 (0)	0 (0)	0 (0)	0 (0)	2 (2)	1 (1)	0 (0)	4 (5)	0 (0)
No, n (%)	86 (99)	91 (100)	99 (100)	96 (100)	80 (100)	90 (98)	68 (99)	92 (100)	79 (95)	93 (100)
Decreased vision										
Yes, n (%)	48 (49)	57 (54)	59 (55)	41 (40)	45 (47)	44 (44)	32 (45)	41 (41)	37 (41)	37 (37)
No, n (%)	50 (51)	48 (46)	48 (45)	62 (60)	50 (53)	55 (56)	39 (55)	59 (59)	53 (59)	64 (63)
Floaters										
Yes, n (%)	19 (21)	16 (16)	9 (9)	17 (18)	15 (18)	14 (15)	8 (12)	15 (16)	10 (13)	16 (17)
No, n (%)	72 (79)	81 (84)	92 (91)	80 (82)	67 (82)	80 (85)	61 (88)	78 (84)	70 (88)	78 (83)
Nuclide										
Ru ¹⁰⁶ , n (%)	21 (17)	10 (8)	72 (58)	120 (97)	119 (96)	121 (98)	120 (97)	121 (98)	120 (97)	119 (97)
I ¹²⁵ , n (%)	102 (83)	114 (92)	52 (42)	4 (3)	5 (4)	3 (2)	4 (3)	3 (2)	4 (3)	4 (3)
TTT before brachytherapy										
Yes, n (%)	3 (3)	2 (2)	23 (19)	21 (18)	25 (22)	25 (23)	7 (9)	30 (26)	23 (22)	25 (23)
No, n (%)	111 (97)	121 (98)	96 (81)	96 (82)	87 (78)	86 (77)	67 (91)	86 (74)	82 (78)	83 (77)

AJCC, American Joint Committee on Cancer; Gy, Gray; TTT, transpupillary thermal treatment.

Larger tumours were more frequent in the lower dose deciles, which is not surprising due to the limiting factor of a maximum scleral dose of 1500 Gy. As expected, the type of radioisotope used (ruthenium-106 or iodine-125, Pearson χ^2 $p < 0.001$) was not equally distributed. The lower dose deciles were over-represented by iodine-125, since the prescription dose is 80 Gy compared with 100 Gy for ruthenium-106. Treatment with TTT before brachytherapy (Pearson χ^2 $p < 0.001$) and pretreatment best-corrected visual acuity (Kruskal-Wallis $p = 0.004$) was not equally distributed across the deciles, but did not show a clear pattern of distribution (table 1).

Dose rate

Descriptive statistics

The average dose rate to the tumour apex plus 1 mm ranged from 0.5 Gy/hour in the first decile to 2.8 Gy/hour in the tenth, and patient characteristics and symptoms were equally distributed across the deciles.

As expected, tumour diameter (Kruskal-Wallis $p < 0.001$), tumour thickness (Kruskal-Wallis $p < 0.001$) and consequently AJCC T category (Pearson χ^2 $p < 0.001$) were not equally distributed across the deciles. Uneven distribution was also seen for

tumour distance to the optic disc (Kruskal-Wallis $p = 0.006$) and tumour distance to the macula (Kruskal-Wallis $p < 0.001$), which were longer in the lower dose and dose rate deciles. Ruthenium-106 was more commonly used in the lowest and highest dose rate deciles, while iodine-125 was used in the middle dose rates deciles (Pearson χ^2 $p < 0.001$). Patients who were treated with TTT before brachytherapy (Pearson χ^2 $p < 0.001$) and pretreatment visual acuity (Kruskal-Wallis $p = 0.001$) were not equally distributed across the deciles, again without a clear pattern of distribution (table 2).

Patient survival

The median follow-up for patients without metastasis was 7.5 years (IQR 5.8). Of all 1238 patients, 462 (37 %) patients were deceased by the end of follow-up. Of the deceased patients, 266 died from melanoma, 111 from other causes and 81 patients from unknown causes. Estimated Kaplan-Meier disease-specific survival revealed a significant difference in melanoma-related mortality between high and low dose and dose rate, respectively ($p = 0.0004$; figure 1). This comparison did not take into account differences in tumour size or location. In further analysis, tumour diameter, tumour thickness and tumour distance to

Table 2 Demographics and clinical features of patients, symptoms, tumours and treatment across deciles based on radiation dose rate

Dose rate decile	1 n=124	2 n=124	3 n=124	4 n=124	5 n=124	6 n=124	7 n=124	8 n=124	9 n=123	10 n=123
Mean dose rate apex, Gy (SD)	0.45 (0.08)	0.63 (0.05)	0.80 (0.05)	0.94 (0.05)	1.09 (0.05)	1.27 (0.06)	1.47 (0.06)	1.70 (0.09)	2.04 (0.13)	2.78 (0.40)
Mean patient age, years (SD)	61.9 (13.8)	62.5 (15.2)	66.6 (11.7)	62.6 (13.6)	64.5 (14.6)	63.6 (14.1)	64.5 (12.8)	62.7 (13.5)	62.1 (14.3)	65.0 (14.2)
Median symptom duration, months (IQR)	2.0 (1.5)	2.0 (5.0)	2.0 (3.0)	2.0 (5.0)	2.0 (2.0)	2.0 (5.0)	2.0 (3.5)	2.0 (0)	2.0 (2.0)	2.0 (5.0)
Mean visual acuity (SD)	0.4 (0.4)	0.3 (0.5)	0.4 (0.4)	0.3 (0.4)	0.4 (0.4)	0.4 (0.4)	0.4 (0.4)	0.5 (0.5)	0.4 (0.5)	0.5 (0.5)
Mean tumour diameter, mm (SD)	12.1 (2.8)	12.0 (3.1)	11.8 (3.2)	11.6 (3.5)	11.6 (3.0)	11.2 (3.2)	10.4 (3.5)	10.0 (3.5)	8.6 (3.3)	8.1 (3.0)
Mean tumour thickness, mm (SD)	6.7 (1.2)	6.7 (1.7)	6.7 (2.3)	6.6 (2.4)	6.0 (2.3)	5.2 (2.0)	4.7 (1.9)	4.2 (1.9)	3.2 (1.2)	2.5 (0.6)
Mean distance to macula, mm (SD)	4.4 (3.8)	4.1 (3.6)	4.7 (4.2)	4.0 (3.8)	3.7 (3.6)	3.1 (2.5)	3.0 (3.3)	3.3 (4.2)	2.5 (3.4)	3.5 (4.0)
Mean distance to optic disc, mm (SD)	4.4 (3.5)	4.3 (3.2)	4.8 (4.1)	3.9 (3.5)	4.1 (3.4)	3.5 (2.6)	3.5 (3.0)	3.7 (4.1)	3.0 (2.7)	3.5 (3.7)
AJCC										
T category										
1, n (%)	4 (3)	13 (11)	20 (17)	19 (16)	20 (17)	26 (22)	49 (40)	57 (48)	88 (77)	104 (87)
2, n (%)	75 (63)	69 (58)	60 (52)	57 (48)	61 (51)	69 (58)	53 (43)	42 (36)	21 (18)	14 (12)
3, n (%)	37 (31)	36 (30)	30 (26)	40 (34)	36 (30)	23 (19)	19 (16)	18 (15)	5 (4)	1 (1)
4, n (%)	4 (3)	2 (2)	6 (5)	3 (3)	2 (2)	2 (2)	1 (1)	1 (1)	1 (1)	1 (1)
Laterality										
Right, n (%)	60 (61)	44 (41)	58 (50)	59 (50)	58 (50)	65 (56)	53 (47)	56 (50)	56 (53)	37 (39)
Left, n (%)	38 (39)	64 (59)	57 (50)	60 (50)	58 (50)	52 (44)	59 (53)	56 (50)	50 (47)	57 (61)
Gender										
Male, n (%)	57 (46)	58 (48)	62 (50)	56 (45)	66 (53)	68 (55)	66 (54)	58 (47)	65 (53)	66 (54)
Female, n (%)	67 (54)	64 (52)	62 (50)	68 (55)	58 (47)	56 (45)	57 (46)	66 (53)	58 (47)	57 (46)
Pain before treatment										
Yes, n (%)	0 (0)	2 (2)	1 (1)	1 (1)	2 (2)	0 (0)	1 (1)	1 (1)	0 (0)	0 (0)
No, n (%)	82 (100)	93 (98)	87 (99)	89 (99)	87 (98)	96 (100)	88 (99)	91 (99)	81 (100)	79 (100)
Decreased vision										
Yes, n (%)	47 (51)	49 (51)	45 (46)	53 (53)	52 (50)	44 (42)	40 (40)	42 (42)	36 (40)	33 (40)
No, n (%)	46 (49)	48 (49)	53 (54)	47 (47)	52 (50)	60 (58)	60 (60)	58 (58)	55 (60)	49 (60)
Floaters										
Yes, n (%)	12 (14)	14 (14)	12 (13)	18 (19)	13 (14)	18 (18)	19 (21)	13 (14)	10 (12)	10 (13)
No, n (%)	71 (86)	83 (86)	78 (87)	75 (81)	79 (86)	81 (82)	72 (79)	81 (86)	71 (88)	69 (87)
Nuclide										
Ru ¹⁰⁶ , n (%)	116 (94)	93 (75)	82 (66)	65 (52)	80 (65)	86 (69)	88 (71)	96 (77)	115 (93)	122 (99)
I ¹²⁵ , n (%)	8 (6)	31 (25)	42 (34)	59 (48)	44 (35)	38 (31)	36 (29)	28 (23)	8 (7)	1 (1)
TTT before brachytherapy										
Yes, n (%)	5 (5)	11 (10)	12 (10)	19 (16)	13 (11)	23 (20)	26 (23)	21 (19)	28 (26)	26 (27)
No, n (%)	93 (95)	98 (90)	103 (90)	100 (84)	103 (89)	94 (80)	86 (77)	91 (81)	78 (74)	69 (73)

Gy, Gray; TTT, transpupillary thermal therapy.

the optic disc were entered as covariates in a multivariate Cox regression analysis of dose and dose rate, respectively.

Dose

Of the covariates, tumour diameter (HR 1.1, 95 % CI 1.1 to 1.2, $p < 0.001$) was independently associated with melanoma-related mortality, but not tumour thickness (HR 1.0, 95 % CI 1.0 to 1.1, $p = 0.4$) or tumour distance to the optic disc (HR 1.0, 95 % CI 1.0 to 1.1, $p = 0.09$). A 1 Gy decrease in the apex dose was not associated with melanoma-related mortality (HR 1.0, 95 % CI 1.0 to 1.0, $p = 0.2$). Similarly, each decreased quartile (HR 1.0, 95 % CI 0.8 to 1.1, $p = 0.6$) and each decreased decile (HR 1.0, 95 % CI 0.9 to 1.0, $p = 0.4$) in apex dose were not associated with melanoma-related mortality (figure 2).

Dose rate

Of the covariates, tumour diameter (HR 1.1, 95 % CI 1.1 to 1.2, $p < 0.001$) was independently associated with melanoma-related mortality, but not tumour thickness (HR 1.1, 95 % CI 1.0 to 1.2, $p = 0.07$) or tumour distance to the optic disc (HR 1.1, 95 % CI 1.0 to 1.1, $p = 0.1$). A 1 Gy/hour decrease in the apex dose rate was not associated with melanoma-related mortality (HR 1.1, 95 % CI 0.8 to 1.4, $p = 0.7$). Similarly, each decreased quartile (HR 1.0, 95 % CI 0.9 to 1.2, $p = 0.5$) and each decreased decile

(HR 1.0, 95 % CI 1.0 to 1.1, $p = 0.6$) in apex dose rate were not associated with melanoma-related mortality (figure 3).

DISCUSSION

In this study, we found no significantly increased hazards for choroidal melanoma-related mortality after plaque brachytherapy with decreasing doses between 108.6 and 73.0 Gy, or with decreasing dose rates between 2.8 and 0.5 Gy/h.

Although challengeable by alternative views on the meta-static process, brachytherapy dose and dose rate have at least a hypothetical implication on patient survival.¹⁶ If the radioactive dose and dose rate approaches 0, tumour cells will be left unharmed and eventually spread to the extraocular environment. Conversely, if the dose and dose rate are extremely high, neither tumour cells nor normal ocular cells will survive. Tumour cells that have already been seeded from the eye to form dormant micrometastases in a distant organ, primarily the liver, will naturally not be affected by the treatment.¹³ If micrometastases were exclusively seeded during a limited time period in early tumour development and a continuous seeding was not in effect during further tumour growth, delaying enucleation until a small tumour has grown into a large tumour would not add additional risk for metastasis. This is contradicted by consensus on the great

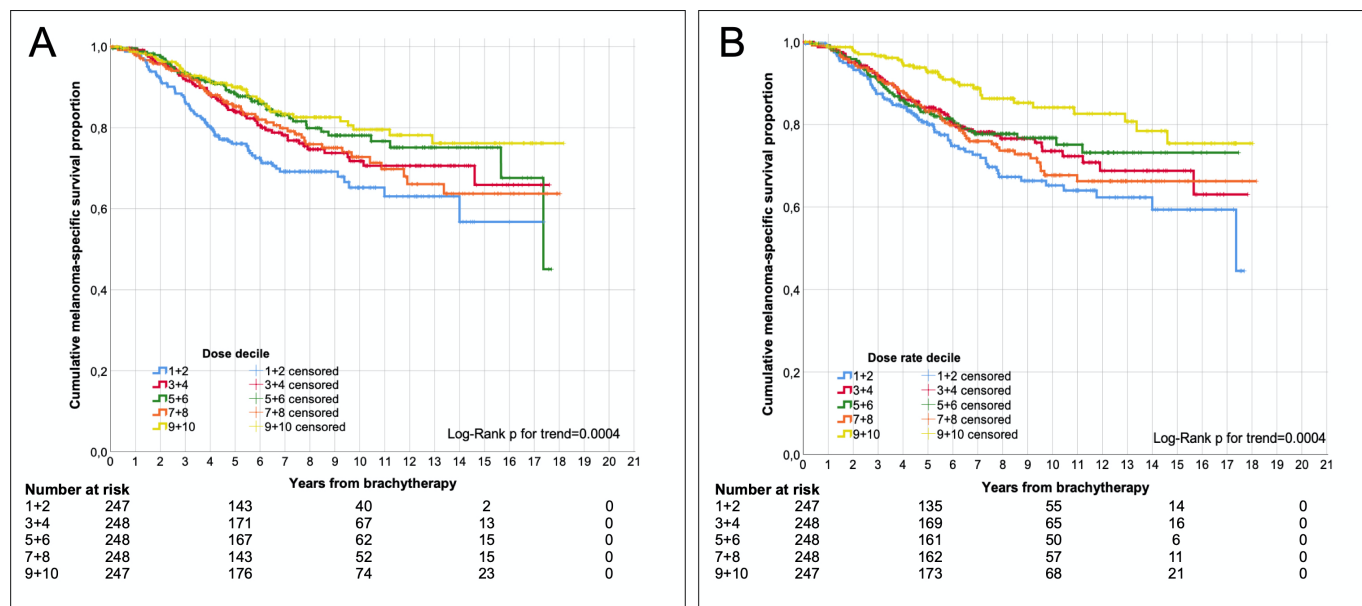


Figure 1 Kaplan-Meier curves, estimated disease-specific survival after brachytherapy with separate (A) dose intervals and (B) dose rate intervals. There was a significant difference between the groups ($p=0.0004$), but further analysis using COX regression model revealed that these differences were due to differences in tumour size rather than dose or dose rate.

importance of tumour size for patient prognosis.^{4 17 18} In the Collaborative Ocular Melanoma Study (COMS), the association between treatment failure and survival was non-significant.⁷ In a multicentre study by the Ophthalmic Oncology Task Force, local recurrence was associated with significantly shorter metastasis-free survival.¹⁹ This may however be a marker for aggressive disease rather than a consequence of postbrachytherapy metastatic seeding.

A possible mechanism behind the fact that small tumours that are left to grow into large tumours increase their metastatic risk, other than continued seeding of micrometastases, may be highlighted by recent findings about tumour cell interaction with the local microenvironment, including the extracellular matrix and immune system: patients who later develop metastases have

primary tumours with significantly higher expression of immune checkpoints T cell Ig and ITIM domain (TIGIT) and Indoleamine 2,3-dioxygenase (IDO).^{20 21} The denser the extracellular matrix and infiltration of macrophages within the tumour, the worse the prognosis.^{22 23} An insufficient radioactive dose or dose rate would consequently leave more surviving cells for such metastasis-enhancing interactions with the microenvironment and for the development of immune evasion. Regardless of the actual mechanism, one can assume that there is a threshold in radioactive dose and dose rate somewhere between 0 and infinity, at which a critical mass of cells survive brachytherapy and the metastatic risk increases. But as our results indicate, this threshold is not within the range of doses and dose rates normally used with therapeutic intent.

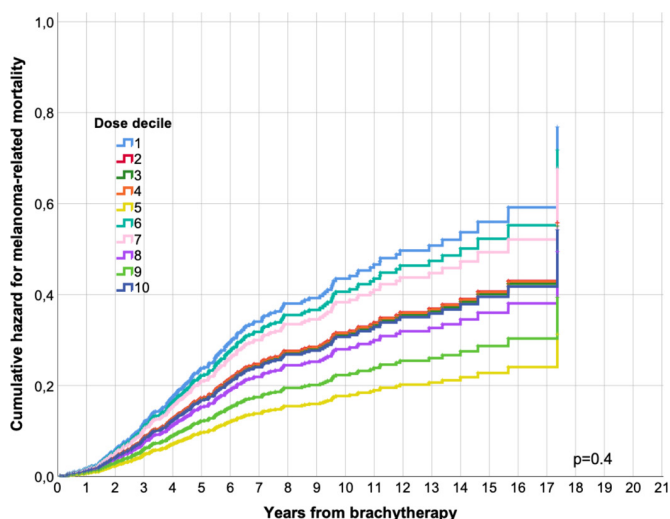


Figure 2 Multivariate COX regression cumulative hazard for melanoma-related mortality at mean of tumour thickness, diameter and distance to the optic disc, across dose deciles (HR 1.0, 95% CI 0.9 to 1.0, $p=0.4$).

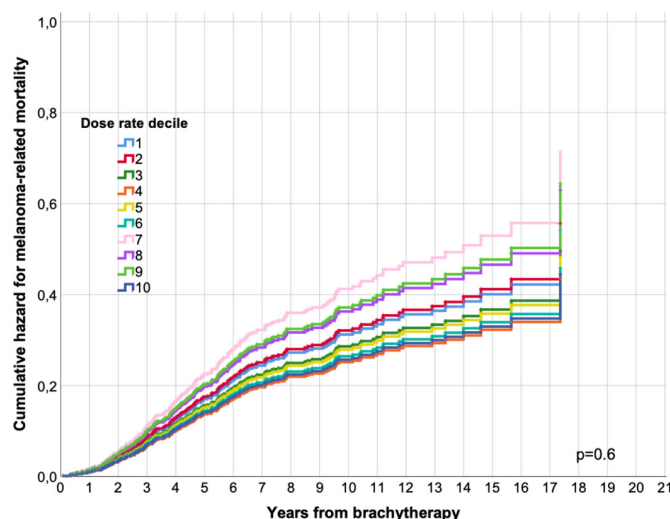


Figure 3 Multivariate COX regression cumulative hazard for melanoma-related mortality at mean of tumour thickness, diameter and distance to the optic disc, across dose rate deciles (HR 1.0, 95% CI 1.0 to 1.1, $p=0.6$).

The generalisability of these findings has limitations. First, the type of radioisotope used was not entered as a covariate in the Cox regression analysis, even though it was unequally distributed between the groups compared. This was motivated by previous studies indicating that the type of radioisotope used is not an independent predictor of survival in choroidal melanoma.^{8 24 25} AJCC T category was not entered since it is a direct calculation based on tumour thickness and diameter, both of which were used as covariates in the Cox regression analysis. For similar reasons, tumour distance to the macula was not included as it was considered closely related to the tumour distance to the optic disc, the latter being included in the analysis. Second, our results are based on retrospective non-randomised data which enable influence of undetected prognostic factors. Third, it is possible that significant statistical differences would have been revealed with a larger patient sample.

In conclusion, we propose that our results are viewed in the context of brachytherapy treatment effect versus complications, including radiation retinopathy and neuropathy, secondary glaucoma, vitreous haemorrhage, retinal detachment and scleral melting, all of which become more prevalent as radiation doses and dose rates increase.^{26–28} Our findings support the safety of decreasing dose and dose rates with regards to risk of metastases, and previous research shows a lack of increased local recurrence within similar dose and dose rate intervals.^{6 9} Seen together, one may argue that lower doses and dose rates down to 75 Gy and 0.5 Gy/hour, respectively, would be both safe and effective.

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